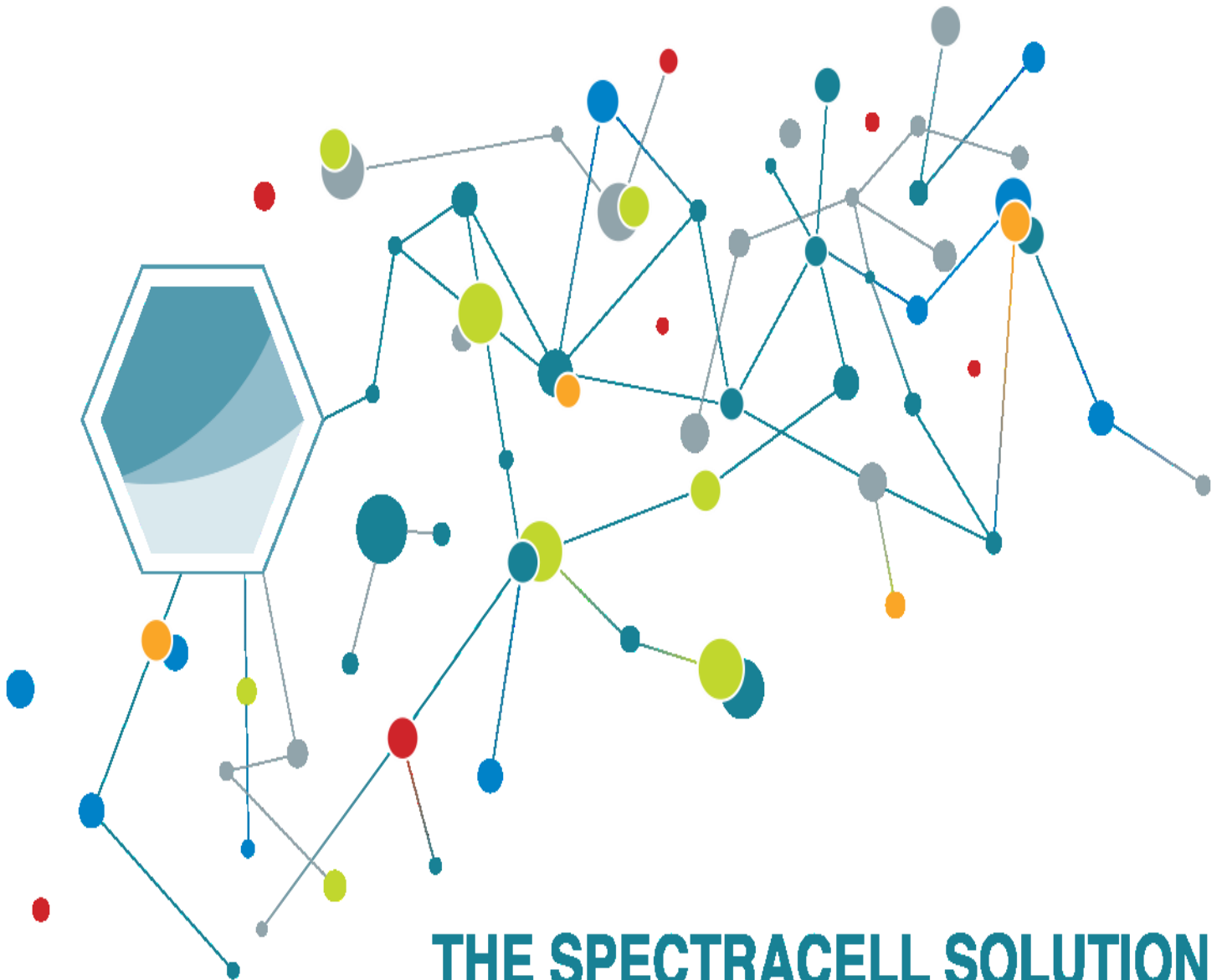




THE SPECTRACELL SOLUTION

Driven by Science. Inspired by You.

Patient: **Craig, William**

Accession ID: 2410050029

Provider: Any Lab Test Now Mesa

Order Status: **Complete**

10401 Town Park Drive, Houston, TX 77072 | 1(800) 227-5227 | support@spectracell.com | www.spectracell.com

PATIENT		SPECIMEN		PROVIDER	
NAME Craig, William	AGE 29	ACCESSION ID 2410050029	DATE COLLECTED 10/04/2024	ACCOUNT ID 74329587	CLIENT NAME Any Lab Test Now Mesa
DOB 3/28/1995	GENDER Male	ORDER ID 1028-00074329587-241005	DATE RECEIVED 10/05/2024	ADDRESS Any Lab Test Now, Mesa	
PATIENT ID 24-279-00022		DATE REPORTED 10/17/2024			
Order Comments:					

Normal Borderline Out of Range

Lipoprotein Particle Numbers					
Tests		In Range	Out of Range	Reference Range	Units
VLDL Particles		84		<85	nmol/L
Total LDL Particles			1014	<900	nmol/L
Non-HDL Particles			1097	<1000	nmol/L
Remnant Lipoprotein			156	<150	nmol/L
Dense LDL III		158		<300	nmol/L
Dense LDL IV		74		<100	nmol/L
Total HDL Particles		8678		>7000	nmol/L
Buoyant HDL 2b		2180		>1500	nmol/L

Lipid Panel					
Tests		In Range	Out of Range	Reference Range	Units
Total Cholesterol			208	<200	mg/dL
Triglycerides		101		<150	mg/dL
HDL			49	>40	mg/dL
LDL			146	40-130	mg/dL
Non-HDL Cholesterol			159	<160	mg/dL

Vascular Inflammation					
Tests		In Range	Out of Range	Reference Range	Units
Insulin		8.6		<24.9	μIU/mL
*Please note change in reference range. Previously <21 ulU/mL, now <24.9 ulU/mL					
hs-CRP		1.18		<3.00	mg/L
Lipoprotein(a)		8.5		<75.0	nmol/L

PATIENT: **Craig, William** PROVIDER: **Any Lab Test Now Mesa** DATE REPORTED: **10/17/2024** ACCESSION ID: **2410050029**

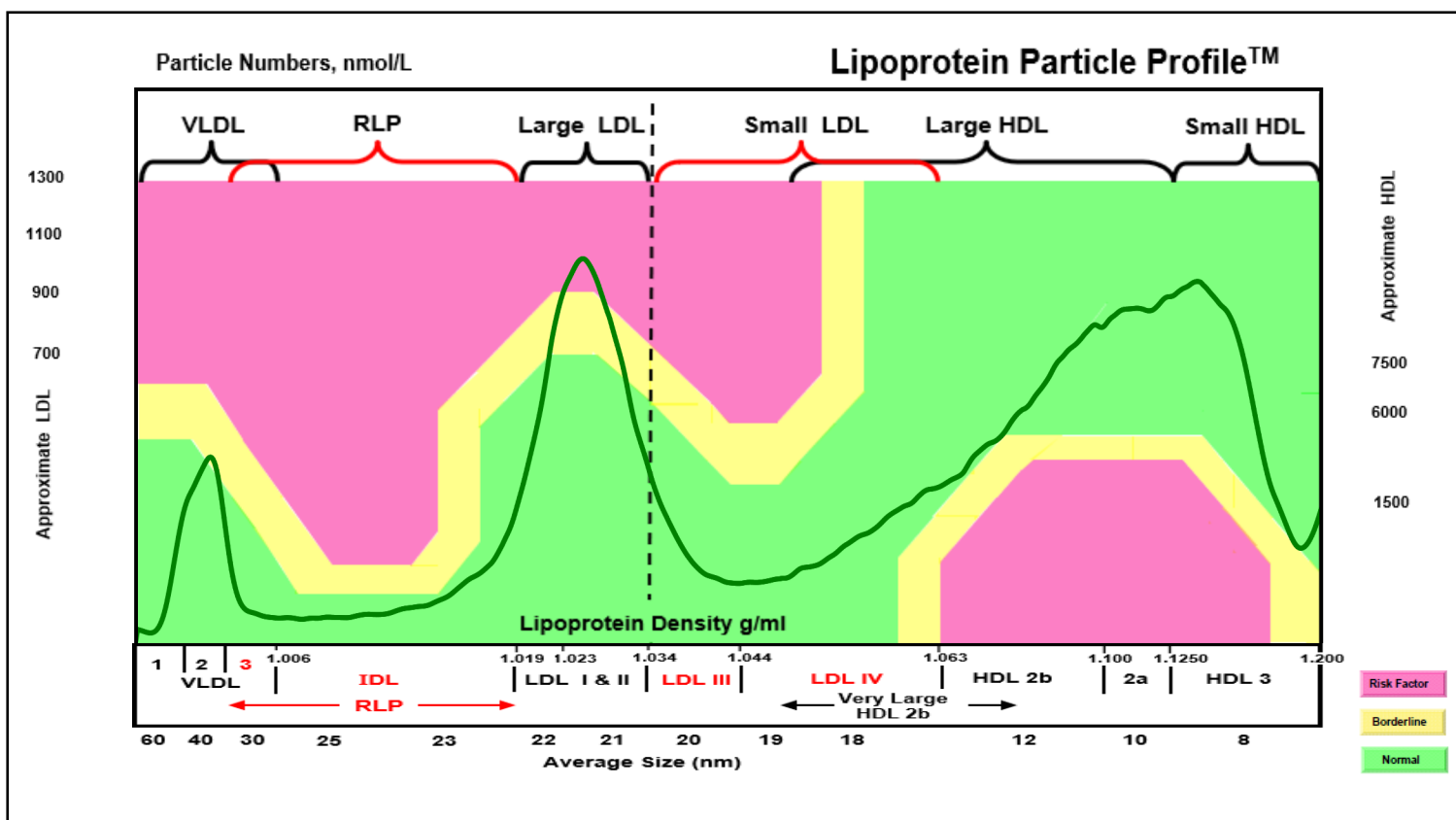
*Please note change in reference range and units of measure. Previously <30 mg/dL, now <75 nmol/L

Apolipoprotein B	30 64 98 132 166 200	105	60-130	mg/dL
Apolipoprotein A1	250 200 150 100 50 0	124	>115	mg/dL
Homocysteine	0 4 9 13 18 22	7.6	<11	μmol/L

Metabolic Syndrome Traits

Tests	In Range	Out of Range	Reference Range	Units
Metabolic Syndrome Traits	0		Zero	

A diagnosis of metabolic syndrome is confirmed if any three of the following traits exist in a patient: (1) high triglycerides [>150 mg/dL]*; (2) low HDL [<40 mg/dL in men, <50 mg/dL in women]*; (3) elevated small dense LDL III and LDL IV [>400 nmol/L]*; (4) high fasting glucose [>100 mg/dL]; (5) high blood pressure [$>130/85$]; (6) high waist circumference [>40 inches in men, >35 inches in women]. *Included in this section of report. Clinician must determine traits (4), (5), (6).



PATIENT: Craig, William

PROVIDER: Any Lab Test Now Mesa

DATE REPORTED: 10/17/2024

ACCESSION ID: 2410050029

Lipoprotein Particle Profile (Component Summaries)

This information is provided for educational purposes.

Lipoprotein Particle Numbers – Lipoproteins are ball-shaped proteins in the blood that transport fats (lipids) throughout the body. The fact that lipoproteins – not the cholesterol that is carried within them – causes cardiovascular disease by penetrating the endothelial lining of the arteries, becoming oxidized and contributing to arterial plaque, has been well established. Further, the most effective treatment will depend on which lipoproteins are elevated, so measuring lipoprotein particle numbers enables a clinician to (1) determine accurately the level of cardiometabolic risk and (2) how best to treat it.

Remnant Lipoprotein (RLP) – This highly atherogenic lipoprotein causes platelet aggregation and impairs vascular relaxation. Unlike other LDL particles which have to be oxidized before they are taken into the arterial intima by macrophage cells, RLP can contribute to plaque buildup even when not oxidized. Foam cells (the sticky contributors to arterial plaque) contains high levels of RLP. Treatment with omega 3 fatty acids can be efficacious.

Dense LDL III and LDL IV – These lipoproteins are small and can thus more easily penetrate and damage the lining of the arteries due to their size, causing plaque and atherosclerosis. They are highly correlated to cardiovascular disease.

HDL2b – This is a protective lipoprotein that indicates how well cholesterol is being cleared by the liver (reverse cholesterol transport system). HDL is made in the liver as HDL3 and as it travels through the body accumulating cholesterol it becomes the larger and lipid-enriched HDL2b. It positively correlates with heart health.

Lipid Panel – The lipid panel measures cholesterol, not lipoproteins (which carry cholesterol). Although directly measuring the actual number of lipoproteins (versus the amount of cholesterol inside them) is widely recognized as a superior tool in assessing cardiometabolic health, clinicians and patients tend to be familiar with a standard lipid panel and its historical use. It is important to note that half of all people who have a heart attack will have cholesterol values that fall in the normal range. Thus, the lipid panel is most useful when viewed in the context of other biomarkers, particularly lipoprotein particle numbers. Elevated triglycerides and low HDL-cholesterol are highly correlated to metabolic syndrome and increase the risk of heart disease significantly.

Vascular Inflammation – Cardiovascular disease is generally considered an inflammatory process and the analytes included here are important determinants of cardiometabolic risk, particularly with respect to vascular inflammation.

Insulin – Insulin is a hormone made by beta cells (β -cells) in the pancreas and secreted in response to elevated blood sugar. Its main function is to regulate plasma glucose levels within a narrow range and is correlated to the efficiency with which a person can metabolize carbohydrates. If one becomes de-sensitized to the action of insulin (insulin resistant), more is needed to achieve adequate glucose-lowering effects, thus altering metabolism to favor fat storage over efficient energy production. High fasting insulin indicates insulin resistance and possible pre-diabetes. Stimulatory hormones (i.e. adrenaline, cortisol) can also raise insulin levels. Test performed by Labcorp, 7207 North Gessner, Houston, TX 77040, CLIA # 45D0674994. Lab Director: Kyle Eskue

hs-CRP – High Sensitivity C-reactive Protein (hs-CRP) is an acute phase protein that reflects the presence of inflammation in the body. High CRP, regardless of cause, is strongly correlated to the risk of sudden cardiac death and low-grade chronic systemic inflammation raises the risk of metabolic syndrome, heart disease, diabetes and other degenerative diseases.

Lipoprotein(a) – This unique lipoprotein is particularly dangerous because it inhibits the formation of plasmin which is an enzyme that dissolves blood clots. High levels of Lp(a) are strongly linked to thrombosis significantly raising the risk of blood clots and associated cardiac events. It can also penetrate the arterial lining, become oxidized and build plaque, thus contributing to atherosclerosis independent of its thrombotic potential. Test performed by Labcorp, 7207 North Gessner, Houston, TX 77040, CLIA # 45D0674994. Lab Director: Kyle Eskue

Apolipoprotein B – ApoB100 is a protein produced in the liver that attached to the surface of all low-density lipoproteins (LDL), regardless of type. Every molecule of VLDL, RLP, Lp(a) and LDL has exactly one, and only one apoB100 molecule attached to it and thus, apoB reflects the level of atherogenic lipoproteins in the blood.

Apolipoprotein A1 – ApoA1 is a protein that is attached to the surface of all high-density lipoproteins (HDL) and is thus reflective of the amount of protective lipoproteins in the blood. It facilitates the removal of fats (cholesterol) from arterial walls by enabling its transport back to the liver for eventual excretion. Like HDL, low levels raise risk of heart disease.

Homocysteine – A metabolic intermediate, this protein is dangerous at high levels because it indicates poor methylation (detoxification) ability. Homocysteine will also act as an arterial abrasive, physically damaging the endothelial lining of blood vessels. High levels are strongly linked to kidney and heart disease, stroke and dementia.

PATIENT: **Craig, William**

PROVIDER: **Any Lab Test Now Mesa**

DATE REPORTED: **10/17/2024**

ACCESSION ID: **2410050029**